TEXORA · LECTURE SHEET

Mechanical Properties of Materials

How materials answer force: the tensile test and its vocabulary — stiffness, yield, strength, ductility, toughness — plus hardness, impact resistance, fatigue and creep, the four ways real components quietly fail.

ME-1207 · ENGINEERING MATERIALS · TOPIC 2 · LEVEL 1 TERM 2

1. The tensile test — a material's biography

A standard specimen is pulled slowly while load and extension are recorded. From them come engineering stress $\sigma = P/A_0$ and strain $\varepsilon = \Delta L/L_0$, and the σ–ε curve tells the whole story:

PROPERTY	MEANING / WHERE READ
Young's modulus E	Slope of the elastic line; stiffness. Steel ≈ 200 GPa, aluminium ≈ 70, polymers 0.1–5 — bonding made visible (Topic 1)
Proportional limit / elastic limit	End of the recoverable region; beyond it, strain is permanent
Yield strength σ_y	Onset of plastic flow — for design, the practical ceiling (mild steel shows a sharp yield point; others use the 0.2% proof stress)
Ultimate tensile strength (UTS)	The curve's peak; necking begins here in ductile metals
Ductility	% elongation = $(L_f - L_0)/L_0 \times 100$ and % reduction of area at fracture — how much warning before failure
Resilience / toughness	Area under the curve: to σ_y (resilience — spring energy, ideal $\sigma_y^2/2E$) or to fracture (toughness — total energy absorbed)

WORKED EXAMPLE

Steel rod, d = 10 mm (A = 78.5 mm²), load 15 kN: σ = 15,000/78.5 = **191 MPa**. Over a 200 mm gauge with E = 200 GPa the elastic stretch is $\Delta L = \sigma L/E = 191 \times 10^6 \times 0.2/(200 \times 10^9)$ = **0.19 mm** — steel is strong *and* stiff, moving little under big loads.

Ductile versus brittle fracture ends the story: ductile metals neck and tear at ≈ 45° into a cup-and-cone surface after absorbing much energy; brittle materials (cast iron, ceramics, steel below its ductile–

brittle transition temperature) snap flat with almost no deformation — the failure mode behind the Liberty ships that cracked in cold seas, and the reason BCC steels are notch-tested for winter service.

2. Hardness

Hardness is resistance to local indentation — a quick, nearly non-destructive proxy for strength and wear resistance. Brinell presses a hardened ball (load P, ball diameter D, dent diameter d):

$$HB = 2P / [\pi D(D - \sqrt{(D^2 - d^2)})]$$

With P = 3,000 kgf, D = 10 mm and a measured dent d = 4.0 mm: HB = 6,000/[π × 10 × (10 − √84)] = 6,000/26.2 ≈ **229** — a medium steel. Rockwell (depth-reading, fast, several scales) rules the factory floor; Vickers (diamond pyramid) spans soft to hardest and down to microhardness of coatings and fibres. As a rule of thumb for steels, UTS (MPa) ≈ 3.45 × HB — hardness numbers quietly double as strength numbers.

3. Impact, fatigue and creep — failure without overload

- **Impact:** the Charpy test swings a pendulum through a notched specimen; the energy absorbed (height lost) grades notch toughness and reveals the ductile–brittle transition temperature.
- **Fatigue:** repeated cycles nucleate a crack at a surface flaw or stress concentration and grow it invisibly until the remaining section fails — at stresses far below σ_y. The S–N curve (stress vs cycles to failure) for steels flattens to an **endurance limit** ≈ 0.5 × UTS below which life is effectively infinite; aluminium alloys have no such plateau and must be designed for a stated life. Rotating shafts, loom parts and springs live or die by fatigue.
- **Creep:** under constant load at high temperature (above ≈ 0.4 T_m in kelvin), materials stretch slowly through three stages — primary (decelerating), secondary (steady — the design rate), tertiary (accelerating to rupture). Boiler tubes, turbine blades and even solder joints creep; the countermeasures are alloy choice (solid-solution and precipitate strengtheners that pin dislocations) and respecting temperature limits.

Threading all three: real design rarely fails by simple overload. It fails by accumulated insult — impacts, cycles, time at temperature — acting on defects. Topic 5’s testing methods exist to find those defects before service does.

4. Key points

- σ = P/A, ε = ΔL/L; E from the elastic slope; σ_y is the design ceiling; UTS the curve’s peak.

- Ductility = % elongation + % reduction of area; toughness = whole area under σ–ε; resilience = area to yield.
- Hardness (Brinell/Rockwell/Vickers) proxies strength: UTS ≈ 3.45 × HB for steels.
- Charpy grades notch toughness and the ductile–brittle transition; BCC steels are vulnerable in the cold.
- Fatigue: S–N curve, steel endurance limit ≈ 0.5 UTS, aluminium none. Creep: three stages above ≈ 0.4 T_m.

5. Practice

Answer in your own words; brief model answers follow.

1. An aluminium rod (d = 12 mm, gauge length 250 mm, E = 70 GPa) carries 20 kN in tension. Find the stress and the elastic extension.
2. A Brinell test with P = 1,500 kgf and D = 10 mm leaves a dent of 3.5 mm. Compute HB and estimate the steel's UTS.
3. A tensile specimen of 50 mm gauge length and 10 mm diameter breaks at 62 mm length with a neck diameter of 7 mm. Find % elongation and % reduction of area, and classify the material's ductility.
4. A machine bolt, tightened well below its yield load, snaps after some months of vibration. Name the failure mechanism and two design changes that fight it.
5. Why are creep data essential for a steam pipe designed for 30 years at 550 °C, when its short-time tensile strength at that temperature looks comfortable?
6. Two steels have identical UTS; one shows a Charpy transition at −10 °C, the other at −60 °C. Which do you choose for a structure in a Siberian winter, and why?

Answer key

1. A = π/4 × 12² = 113.1 mm². σ = 20,000/113.1 = 177 MPa. ΔL = σL/E = 177 × 10^6 × 0.25/(70 × 10^9) = 6.3 × 10^{-4} m = 0.63 mm — over three times the stretch of the steel example, as E is about one third.
2. HB = 2 × 1,500/[π × 10 × (10 − √(100 − 12.25))] = 3,000/[31.42 × (10 − 9.368)] = 3,000/19.85 ≈ 151. Estimated UTS ≈ 3.45 × 151 ≈ 520 MPa — a medium-carbon steel in the normalised condition, roughly.
3. % elongation = (62 − 50)/50 × 100 = 24%. Areas: A_0 = 78.5 mm², A_f = π/4 × 7² = 38.5 mm²; % RA = (78.5 − 38.5)/78.5 × 100 = 51%. Both figures are high — a ductile material (mild-steel-like) that warns before it fails.
4. Fatigue. Fight it by (1) removing stress concentrations — generous fillets, no sharp threads run-out, polished surfaces, shot peening to add compressive surface stress; and (2) keeping the cyclic stress

below the endurance limit — larger section, better alignment to kill bending cycles.

5. Because at 550 °C steel creeps: given years, the pipe wall thins steadily through secondary creep and enters tertiary creep and rupture at stresses far below the short-time UTS. Design must use creep-rupture data (stress for 10^5-hour life, say), not the quick tensile test — time itself is a load at high temperature.
6. The −60 °C steel. Below its transition temperature a BCC steel turns notch-brittle — a small flaw plus a shock load can run a crack through the whole section at low absorbed energy. Siberian service (−40 °C and below) sits under the first steel's transition but above the second's, so only the second remains safely ductile.
